

Variational Methods: Minimization and Minimax Principles

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Abstract

These notes develop a single path from weak formulations and the direct method to constrained minimization, compactness recovery on unbounded domains, natural constraints, deformation theory, mountain-pass and symmetric minimax methods, and quasilinear applications. Repeated material in the source slides has been consolidated, notation has been standardized, and abbreviated proofs have been completed where the missing steps are essential to the logical flow.

1 Weak solutions and the direct method

1.1 Weak derivatives and variational structure

Variational methods apply to differential equations whose weak solutions are critical points of an energy functional. A finite-dimensional model is

$$6x^5 + 4x^3 - 100 = 0,$$

the critical-point equation for $I(x) = x^6 + x^4 - 100x$. Since I is coercive, it has a global minimizer, hence at least one critical point.

For the boundary-value problem

$$-u''(t) = f(t) \quad (0 < t < 1), \quad u(0) = u(1) = 0, \quad (1.1)$$

integration by parts gives

$$\int_0^1 (u' \varphi' - f \varphi) dt = 0 \quad \text{for every } \varphi \in C_c^\infty(0, 1). \quad (1.2)$$

Definition 1.1 (Weak derivative). Let $U, V \in L_{\text{loc}}^1(0, 1)$. We call V the weak derivative of U , and write $U' = V$, if

$$\int_0^1 U \varphi' dt = - \int_0^1 V \varphi dt \quad \text{for every } \varphi \in C_c^\infty(0, 1).$$

Definition 1.2 (Weak solution). A function $u \in H_0^1(0, 1)$ is a weak solution of (1.1) if it satisfies (1.2). Here $H_0^1(0, 1)$ is the closure of $C_c^\infty(0, 1)$ under

$$\|u\|_{H^1}^2 = \int_0^1 (|u'|^2 + |u|^2) dt.$$

If f is continuous, (1.2) says that u' has weak derivative $-f$. One-dimensional regularity then recovers $u'' = -f$, so the weak and classical notions agree. The same principle holds for elliptic PDEs under the usual regularity assumptions.

Let $\Omega \subset \mathbb{R}^N$ be a bounded domain. A weak solution of

$$-\Delta u = f(x) \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega \quad (1.3)$$

is a function $u \in H_0^1(\Omega)$ satisfying

$$\int_\Omega \nabla u \cdot \nabla \varphi dx = \int_\Omega f \varphi dx \quad \text{for every } \varphi \in H_0^1(\Omega). \quad (1.4)$$

We use

$$H_0^1(\Omega) = \overline{C_c^\infty(\Omega)}^{H^1}, \quad (u, v)_{H^1} = \int_\Omega (\nabla u \cdot \nabla v + uv) dx.$$

Definition 1.3 (Fréchet derivative). Let E be a Banach space and $I : E \rightarrow \mathbb{R}$. The functional I is Fréchet differentiable at u if there is $A \in E^*$ such that

$$\lim_{\|h\| \rightarrow 0} \frac{|I(u+h) - I(u) - \langle A, h \rangle|}{\|h\|} = 0.$$

We write $I'(u) = A$. If $I' : E \rightarrow E^*$ is continuous, then $I \in C^1(E, \mathbb{R})$. A point u is critical when $I'(u) = 0$.

For $f \in L^2(\Omega)$ define

$$I(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2 dx - \int_{\Omega} f u dx, \quad u \in H_0^1(\Omega). \quad (1.5)$$

Computing the directional derivative first and then checking the remainder gives

$$\langle I'(u), \varphi \rangle = \int_{\Omega} \nabla u \cdot \nabla \varphi dx - \int_{\Omega} f \varphi dx.$$

Moreover,

$$\|I'(u) - I'(v)\|_{(H_0^1)^*} \leq \|u - v\|_{H_0^1},$$

so $I \in C^1$. Thus the weak solutions of (1.3) are exactly the critical points of (1.5). With elliptic regularity,

$$\text{classical solutions} \iff \text{weak solutions} \iff \text{critical points of } I.$$

1.2 Poincaré and Sobolev inequalities

If Ω is bounded in one coordinate direction, the Poincaré inequality gives a constant $C_P > 0$ such that

$$\int_{\Omega} |u|^2 dx \leq C_P \int_{\Omega} |\nabla u|^2 dx \quad (u \in H_0^1(\Omega)). \quad (1.6)$$

Consequently $\|u\| = \|\nabla u\|_{L^2}$ is an equivalent norm on $H_0^1(\Omega)$. For instance, when Ω lies between the hyperplanes $x_1 = 0$ and $x_1 = 1$, the fundamental theorem of calculus and Cauchy–Schwarz applied along the x_1 direction yield (1.6) first for smooth functions; density gives the general result.

Set

$$2^* = \begin{cases} \frac{2N}{N-2}, & N \geq 3, \\ \infty, & N = 1, 2. \end{cases}$$

The Sobolev embedding $H_0^1(\Omega) \hookrightarrow L^p(\Omega)$ is continuous for $2 \leq p \leq 2^*$, and compact when Ω is bounded and $2 \leq p < 2^*$. Thus a bounded sequence in $H_0^1(\Omega)$ has a subsequence converging strongly in every subcritical L^p space.

1.3 Weak lower semicontinuity and the direct method

Definition 1.4 (Lower semicontinuity). A functional $I : E \rightarrow \mathbb{R} \cup \{+\infty\}$ is lower semicontinuous at u if

$$u_n \rightarrow u \text{ in } E \implies I(u) \leq \liminf_{n \rightarrow \infty} I(u_n).$$

Weak lower semicontinuity is the same statement with strong convergence replaced by weak convergence.

Definition 1.5. A functional $I : E \rightarrow \mathbb{R} \cup \{+\infty\}$ is weakly lower semicontinuous if

$$u_n \rightharpoonup u \text{ in } E \implies I(u) \leq \liminf_{n \rightarrow \infty} I(u_n).$$

It is coercive on $M \subset E$ if $I(u) \rightarrow +\infty$ whenever $u \in M$ and $\|u\| \rightarrow \infty$.

The squared norm on a Hilbert space is weakly lower semicontinuous. Indeed, if $u_n \rightharpoonup u$, then

$$\|u_n\|^2 = \|u_n - u\|^2 + \|u\|^2 + 2(u_n - u, u),$$

and the last term tends to zero. We also use the fact that every bounded sequence in a reflexive Banach space has a weakly convergent subsequence.

Theorem 1.6 (Direct method). *Let E be reflexive, let $M \subset E$ be nonempty and weakly sequentially closed, and let $I : M \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper, coercive, and weakly lower semicontinuous. Then I is bounded below and attains its infimum on M .*

Proof. Choose a minimizing sequence $u_n \in M$. Coercivity makes (u_n) bounded. After passage to a subsequence, $u_n \rightharpoonup u$ in E . Since M is weakly closed, $u \in M$, and weak lower semicontinuity gives

$$\inf_M I \leq I(u) \leq \liminf I(u_n) = \inf_M I.$$

□

For the linear energy (1.5), Poincaré and Cauchy–Schwarz imply

$$I(u) \geq \frac{1}{2} \|u\|^2 - C \|f\|_{L^2} \|u\|,$$

so I is coercive and bounded below. The quadratic term is weakly lower semicontinuous, while $u \mapsto \int_{\Omega} f u$ is a bounded linear functional and therefore weakly continuous. Theorem 1.6 gives a minimizer, which is a critical point and hence a weak solution of (1.3).

Exercise 1.7. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and bounded, and let $F(t) = \int_0^t f(s) ds$. Show that

$$-\Delta u = f(u) \text{ in } \Omega, \text{quad} u = 0 \text{ on } \partial\Omega$$

has a weak solution by minimizing $I(u) = \frac{1}{2} \int |\nabla u|^2 - \int F(u)$. If $f(0) = 0$, give a condition ensuring $\inf I < 0$, and hence the existence of a nonzero solution. Discuss the example $f(u) = a \sin u$.

1.4 Lagrange multipliers and the first eigenvalue

Theorem 1.8 (Lagrange multiplier rule). *Let E be a Hilbert space, $I, G \in C^1(E, \mathbb{R})$, and $M = \{u \in E : G(u) = 1\}$. If $u \in M$ minimizes I on M and $G'(u) \neq 0$, then $I'(u) = \lambda G'(u)$ for some $\lambda \in \mathbb{R}$.*

For the Dirichlet eigenvalue problem, minimize

$$J(u) = \int_{\Omega} |\nabla u|^2 dx \quad \text{on} \quad \left\{ u \in H_0^1(\Omega) : \int_{\Omega} u^2 dx = 1 \right\}.$$

Compactness of $H_0^1(\Omega) \hookrightarrow L^2(\Omega)$ yields a minimizer and

$$\lambda_1 = \inf_{\|u\|_2=1} \int_{\Omega} |\nabla u|^2 dx > 0.$$

The multiplier equation is $-\Delta u = \lambda_1 u$. Higher eigenvalues admit minimax characterizations and, apart from the first positive eigenfunction, their eigenfunctions change sign.

2 Constrained minimization on bounded domains

Let $\Omega \subset \mathbb{R}^N$ be bounded and $2 < p < 2^*$. Consider

$$-\Delta u = |u|^{p-2}u \quad \text{in } \Omega, \text{quad} u = 0 \quad \text{on } \partial\Omega. \quad (2.1)$$

The unconstrained energy

$$\Phi(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2 dx - \frac{1}{p} \int_{\Omega} |u|^p dx$$

is unbounded below because $\Phi(tu) \rightarrow -\infty$ as $t \rightarrow \infty$. It is also unbounded above; on $(0, 1)$, the functions $u_k(t) = \sin(k\pi t)$ have bounded L^p norm and gradient energy of order k^2 .

Instead define

$$I(u) = \int_{\Omega} |\nabla u|^2 dx, \text{quad} \mathcal{M} = \left\{ u \in H_0^1(\Omega) : \int_{\Omega} |u|^p dx = 1 \right\}, \quad m = \inf_{\mathcal{M}} I. \quad (2.2)$$

For $G(u) = \int |u|^p$, one has

$$\langle G'(u), \varphi \rangle = p \int_{\Omega} |u|^{p-2} u \varphi dx, \quad \langle G'(u), u \rangle = p \quad (u \in \mathcal{M}).$$

Thus the constraint is regular.

Theorem 2.1 (Subcritical constrained minimizer). *The number m in (2.2) is positive and attained by some $u \in \mathcal{M}$. A minimizer may be chosen positive in Ω . After the rescaling $v = m^{1/(p-2)}u$, the function v is a positive weak solution of (2.1).*

Proof. Sobolev's inequality gives $1 = \|u\|_p^p \leq C \|\nabla u\|_2^p$ on $\mathcal{M}$, so $m > 0$. Let $(u_n) \subset \mathcal{M}$ be minimizing. It is bounded in H_0^1 , hence, after passage to a subsequence,

$$u_n \rightharpoonup u \text{ in } H_0^1(\Omega), \text{quad} u_n \rightarrow u \text{ in } L^p(\Omega).$$

The strong convergence preserves the constraint, and weak lower semicontinuity gives $I(u) = m$. The multiplier rule yields

$$2 \int_{\Omega} \nabla u \cdot \nabla \varphi dx = \lambda p \int_{\Omega} |u|^{p-2} u \varphi dx.$$

Testing with u gives $2m = \lambda p$, so the coefficient on the right is m . Replacing a minimizing sequence by its absolute value shows that u may be chosen nonnegative; the strong maximum principle makes it positive. $\square$

The same argument applies to uniformly elliptic divergence-form operators and to nonlinearities $a(x)|u|^{p-2}u$ when a is bounded above and below by positive constants. It also works for $1 < p < 2$, because on a bounded domain the embedding into L^p remains compact.

2.1 The critical exponent

For $N \geq 3$, let $D^{1,2}(\mathbb{R}^N)$ be the completion of $C_c^\infty(\mathbb{R}^N)$ under $\|u\| = \|\nabla u\|_2$, and define the best Sobolev constant

$$S = \inf \left\{ \int_{\mathbb{R}^N} |\nabla u|^2 dx : u \in D^{1,2}(\mathbb{R}^N), \|u\|_{2^*} = 1 \right\}.$$

The infimum is attained. Up to translations, dilations, and normalization, all positive minimizers are Talenti bubbles

$$U_{\lambda,y}(x) = \lambda^{(N-2)/2} U(\lambda(x-y)), \quad U(x) = \frac{[N(N-2)]^{(N-2)/4}}{(1+|x|^2)^{(N-2)/2}}.$$

They solve $-\Delta U = U^{2^*-1}$ in $\mathbb{R}^N$.

For a bounded proper domain Ω , define the analogous constant using $H_0^1(\Omega)$ and extend functions by zero. Concentrating cut-off bubbles show $S(\Omega) = S$, but equality is not attained. This is the basic loss of compactness at the critical exponent.

Exercise 2.2. Verify directly that the displayed U solves $-\Delta U = U^{2^*-1}$, and check that translation and critical scaling leave the Sobolev quotient invariant.

3 Whole-space problems and loss of compactness

Consider the nonlinear Schrödinger equation

$$-\Delta u + V(x)u = |u|^{p-2}u \quad \text{in } \mathbb{R}^N, \quad u(x) \rightarrow 0 \text{ as } |x| \rightarrow \infty, \quad (3.1)$$

where $2 < p < 2^*$ and

$$V \in C(\mathbb{R}^N), \quad V(x) \geq V_0 > 0. \quad (V_0)$$

Such equations arise from standing waves $\psi(t, x) = e^{i\omega t}u(x)$ of a time-dependent nonlinear Schrödinger equation.

Set

$$E = \left\{ u \in H^1(\mathbb{R}^N) : \int_{\mathbb{R}^N} V(x)u^2 dx < \infty \right\}, \quad \|u\|_E^2 = \int_{\mathbb{R}^N} (|\nabla u|^2 + V(x)u^2) dx,$$

and consider

$$m = \inf_{u \in \mathcal{M}} \|u\|_E^2, \quad \mathcal{M} = \{u \in E : \|u\|_p = 1\}. \quad (3.2)$$

The argument of Theorem 2.1 works as soon as $E \hookrightarrow L^p(\mathbb{R}^N)$ is compact. In general it is not: if $V \equiv 1$ and $w \in C_c^1(B_1)$, the translates $w_n(x) = w(x - y_n)$ with $|y_n| \rightarrow \infty$ have constant H^1 and L^p norms but no strongly convergent L^p subsequence.

3.1 Coercive potentials

Assume first

$$V(x) \rightarrow +\infty \quad \text{as } |x| \rightarrow \infty. \quad (V_1)$$

Theorem 3.1 (Compact embedding for a coercive potential). *Under (V_0) and (V_1) , the embedding $E \hookrightarrow L^p(\mathbb{R}^N)$ is compact for $2 \leq p < 2^*$.*

Proof. Suppose $u_n \rightharpoonup u$ in E and put $v_n = u_n - u$. Local Rellich compactness gives $v_n \rightarrow 0$ in $L^2(B_R)$ for every R . Since (v_n) is bounded in E ,

$$\int_{B_R^c} |v_n|^2 dx \leq \frac{1}{\inf_{B_R^c} V} \int_{B_R^c} V |v_n|^2 dx,$$

which is uniformly small for large R . Hence $v_n \rightarrow 0$ in $L^2(\mathbb{R}^N)$. For $2 < p < 2^*$, choose $\theta \in (0, 1)$ with $1/p = \theta/2 + (1 - \theta)/2^*$ and interpolate:

$$\|v_n\|_p \leq \|v_n\|_2^\theta \|v_n\|_{2^*}^{1-\theta} \rightarrow 0.$$

□

It follows immediately that (3.2) is attained and that (3.1) has a positive weak solution after rescaling the minimizer.

A weaker compactness assumption is the following: for some $r > 0$ and every $b > 0$,

$$\lim_{|y| \rightarrow \infty} |\{x : V(x) \leq b\} \cap B_r(y)| = 0. \quad (V_2)$$

Here $|\cdot|$ denotes Lebesgue measure.

Theorem 3.2. *Under (V_0) and (V_2) , the embedding $E \hookrightarrow L^p(\mathbb{R}^N)$ is compact for $2 \leq p < 2^*$.*

Proof idea. Cover $\mathbb{R}^N$ by balls $B_r(y_i)$ with uniformly bounded overlap. Split each ball into $\{V > b\}$ and $A_b(y_i) = B_r(y_i) \cap \{V \leq b\}$. On the first part,

$$\int_{\{V > b\} \cap B_r(y_i)} v_n^2 \leq b^{-1} \int_{B_r(y_i)} V v_n^2.$$

On $A_b(y_i)$, Hölder and the local Sobolev inequality give

$$\int_{A_b(y_i)} v_n^2 \leq C |A_b(y_i)|^{2/N} \|v_n\|_{H^1(B_r(y_i))}^2.$$

After summation, choose first b large and then the exterior radius large. Local compactness handles the remaining bounded region. Interpolation gives the result for all subcritical p . $\square$

3.2 Two splitting tools

The next lemma distinguishes vanishing from concentration.

Lemma 3.3 (Lions' vanishing lemma). *Let (u_n) be bounded in $H^1(\mathbb{R}^N)$, let $r > 0$, and suppose for some $q \in [2, 2^*]$ that*

$$\sup_{y \in \mathbb{R}^N} \int_{B_r(y)} |u_n|^q dx \longrightarrow 0.$$

Then $u_n \rightarrow 0$ in $L^p(\mathbb{R}^N)$ for every $2 < p < 2^$. If $q = 2^*$, then the conclusion also holds for $p = 2^*$.*

For $q = 2$, cover $\mathbb{R}^N$ by balls with bounded overlap and choose $s \in (2, 2^*)$ so that interpolation on each ball produces

$$\int_{B_r(y)} |u_n|^s \leq C \left(\sup_z \int_{B_r(z)} |u_n|^2 \right)^\alpha \int_{B_r(y)} (|\nabla u_n|^2 + |u_n|^2).$$

Summation gives convergence in L^s ; interpolation covers the remaining subcritical exponents. The cases $q > 2$ reduce to this argument.

Lemma 3.4 (Brezis–Lieb). *Let $1 \leq p < \infty$. If (u_n) is bounded in $L^p(\Omega)$ and $u_n \rightarrow u$ almost everywhere, then*

$$\int_{\Omega} |u_n|^p dx = \int_{\Omega} |u_n - u|^p dx + \int_{\Omega} |u|^p dx + o(1).$$

Proof idea. Use, for every $\varepsilon > 0$,

$$||a + b|^p - |a|^p| \leq \varepsilon |a|^p + C_\varepsilon |b|^p.$$

The positive part of the error after subtracting $\varepsilon |u_n - u|^p$ is dominated by a multiple of $|u|^p$ and converges pointwise to zero. Dominated convergence followed by $\varepsilon \downarrow 0$ completes the proof. $\square$

3.3 Periodic potentials

Assume

$$V \text{ is one-periodic in each coordinate.} \quad (V_3)$$

Then $E = H^1(\mathbb{R}^N)$ with equivalent norms, but compact embedding still fails. Periodicity nevertheless lets us recenter a minimizing sequence without changing its energy.

Theorem 3.5 (Periodic case). *Under (V_0) and (V_3) , equation (3.1) has a positive weak solution in $H^1(\mathbb{R}^N)$.*

Proof. Let $(u_n) \subset \mathcal{M}$ minimize (3.2). Since $\|u_n\|_p = 1$, Lemma 3.3 rules out vanishing: there are $r, \delta > 0$ and $y_n \in \mathbb{R}^N$ such that

$$\int_{B_r(y_n)} |u_n|^p dx \geq \delta.$$

After increasing r , take $y_n \in \mathbb{Z}^N$ and set $v_n(x) = u_n(x + y_n)$. Periodicity makes (v_n) another minimizing sequence. Up to a subsequence,

$$v_n \rightharpoonup v \text{ in } H^1(\mathbb{R}^N), \quad v_n \rightarrow v \text{ in } L_{\text{loc}}^p(\mathbb{R}^N),$$

and the local mass bound gives $v \neq 0$.

Write $\alpha = \|v\|_p^p \in (0, 1]$. If $\alpha < 1$, the Brezis–Lieb lemma and Hilbert-space splitting give

$$\begin{aligned} 1 &= \|v_n - v\|_p^p + \alpha + o(1), \\ m + o(1) &= \|v_n - v\|_E^2 + \|v\|_E^2 + o(1) \\ &\geq m \left(\|v_n - v\|_p^2 + \|v\|_p^2 \right) + o(1). \end{aligned}$$

Letting $n \rightarrow \infty$ yields

$$1 \geq (1 - \alpha)^{2/p} + \alpha^{2/p} > 1,$$

because $2/p < 1$. Hence $\alpha = 1$, so $v \in \mathcal{M}$ and weak lower semicontinuity shows $\|v\|_E^2 = m$. Rescaling gives a nonnegative solution; the maximum principle gives positivity. $\square$

3.4 Potential wells

Assume that

$$V(x) \leq V_\infty := \lim_{|x| \rightarrow \infty} V(x) < \infty, \quad V \not\equiv V_\infty. \quad (V_4)$$

Let m_∞ be the constrained minimum for the constant-potential problem with V_∞ . Theorem 3.5 gives a positive minimizer u_∞ . Since $V < V_\infty$ on a nonempty open set,

$$0 < m \leq \int (|\nabla u_\infty|^2 + V u_\infty^2) < \int (|\nabla u_\infty|^2 + V_\infty u_\infty^2) = m_\infty.$$

Theorem 3.6 (Potential-well case). *Under (V_0) and (V_4) , the infimum m is attained and (3.1) has a positive weak solution.*

Key compactness argument. Let $u_n \rightharpoonup u$ be a minimizing sequence. If $u = 0$, local compactness and $V(x) \rightarrow V_\infty$ imply

$$\int_{\mathbb{R}^N} (V - V_\infty) u_n^2 dx = o(1).$$

Therefore $m + o(1) \geq m_\infty + o(1)$, contradicting $m < m_\infty$. Thus $u \neq 0$. Brezis–Lieb splitting and the same strict concavity argument used in the periodic case show $\|u\|_p = 1$. Hence u is a minimizer. $\square$

For the semiclassical equation

$$-\Delta u + V(\varepsilon x)u = |u|^{p-2}u,$$

a standard well condition is $\liminf_{|x| \rightarrow \infty} V(x) > \inf_{\mathbb{R}^N} V(x)$. For all sufficiently small $\varepsilon > 0$, comparison with the limiting autonomous problem produces a positive ground state concentrating near the minimum set of V .

3.5 Concentration–compactness trichotomy

Let $\mu_n \geq 0$ in $L^1(\mathbb{R}^N)$ with $\int \mu_n = 1$ and define

$$Q_n(r) = \sup_{y \in \mathbb{R}^N} \int_{B_r(y)} \mu_n \, dx.$$

After passage to a subsequence, exactly one of the following occurs.

- (i) *Vanishing*: for every $R > 0$, $\sup_y \int_{B_R(y)} \mu_n \rightarrow 0$.
- (ii) *Compactness up to translation*: for every $\varepsilon > 0$ there are $R > 0$ and y_n such that $\liminf_n \int_{B_R(y_n)} \mu_n \geq 1 - \varepsilon$.
- (iii) *Dichotomy*: the mass splits asymptotically into two parts of sizes α and $1 - \alpha$ for some $0 < \alpha < 1$, with the two parts separated by a distance tending to infinity.

The periodic proof rules out vanishing by Lions' lemma and dichotomy by the strict inequality $(1 - \alpha)^{2/p} + \alpha^{2/p} > 1$. This is a concrete instance of the general concentration–compactness principle.

4 Symmetric criticality and symmetry breaking

4.1 Natural constraints and group actions

Let $I \in C^1(E, \mathbb{R})$. A submanifold $M \subset E$ is a *natural constraint* if every critical point of $I|_M$ is a critical point of I on the full space.

Definition 4.1. A topological group G acts linearly on a normed space E if $(g, u) \mapsto gu$ is continuous, $1u = u$, $(gh)u = g(hu)$, and each map $u \mapsto gu$ is linear. The action is isometric if $\|gu\| = \|u\|$. Set

$$\text{Fix}(G) = \{u \in E : gu = u \text{ for all } g \in G\}.$$

A functional is G -invariant if $I(gu) = I(u)$. A map $T : E \rightarrow E$ is G -equivariant if $T(gu) = gT(u)$.

Theorem 4.2 (Palais' principle of symmetric criticality). *Let G act isometrically on a Hilbert space E , and let $I \in C^1(E, \mathbb{R})$ be G -invariant. If $u \in \text{Fix}(G)$ is critical for $I|_{\text{Fix}(G)}$, then $I'(u) = 0$ in E^* .*

Proof. Invariance and differentiation give

$$\langle I'(gu), \varphi \rangle = \langle I'(u), g^{-1}\varphi \rangle.$$

Using the Riesz representation of I' and the isometry of the action,

$$\nabla I(gu) = g\nabla I(u).$$

For $u \in \text{Fix}(G)$ this implies $\nabla I(u) \in \text{Fix}(G)$. Since the derivative of $I|_{\text{Fix}(G)}$ vanishes, testing in the direction $\nabla I(u)$ gives $\|\nabla I(u)\|^2 = 0$. $\square$

If G is a closed subgroup of $O(N)$ and Ω is G -invariant, define $(gu)(x) = u(g^{-1}x)$. This is an isometric action on $H_0^1(\Omega)$, and the fixed-point space consists of G -invariant functions. For $G = O(N)$, these are the radial functions.

4.2 Radial compactness

Suppose $V(x) = V(|x|)$ satisfies (V_0) . Restrict the constrained problem (3.2) to

$$E_r = \{u \in E : u \text{ is radial}\}, \quad \mathcal{M}_r = \{u \in E_r : \|u\|_p = 1\}.$$

Theorem 4.3. *If $N \geq 2$ and $2 < p < 2^*$, the radial constrained minimum is attained. Consequently (3.1) has a positive radial weak solution.*

Compactness mechanism. Let $u_n \in \mathcal{M}_r$ be minimizing and $u_n \rightharpoonup u$. If $u = 0$, Lions' lemma provides y_n with $|y_n| \rightarrow \infty$ and fixed positive L^p mass in $B_r(y_n)$. For any k , when n is large one can rotate $B_r(y_n)$ to k pairwise disjoint balls. Radiality makes the mass in every rotated ball the same, so

$$1 = \|u_n\|_p^p \geq k\delta,$$

a contradiction for large k . Hence $u \neq 0$, and the splitting argument of Theorem 3.5 gives $\|u\|_p = 1$. Symmetric criticality then lifts the restricted critical point to the full space. $\square$

The same argument proves the compact embedding $E_r \hookrightarrow L^p(\mathbb{R}^N)$ for $N \geq 2$ and $2 < p < 2^*$. In contrast, if $V(x) \geq V_\infty = \lim_{|x| \rightarrow \infty} V(x)$ and the inequality is strict somewhere, the unrestricted constrained infimum need not be attained: mass can escape to infinity toward the lower limiting potential.

4.3 Symmetry breaking on expanding annuli

Let

$$\Omega_a = \{x \in \mathbb{R}^N : a < |x| < a + 1\}, \quad N \geq 2,$$

and consider

$$-\Delta u + u = u^{p-1}, \quad u > 0 \text{ in } \Omega_a, \quad u = 0 \text{ on } \partial\Omega_a. \quad (4.1)$$

Let $m_{r,a}$ be the constrained minimum over radial functions and m_a the minimum over all of $H_0^1(\Omega_a)$. Both are attained. The radial minimizer gives a solution by Theorem 4.2. As $a \rightarrow \infty$,

$$m_{r,a} \rightarrow \infty, \quad \text{whereas} \quad \sup_{a \geq 1} m_a < \infty.$$

The first fact reflects the growing surface area forced on a radial profile; the second follows by translating a fixed compactly supported test function inside the annulus. Therefore $m_a < m_{r,a}$ for large a , so the unrestricted minimizer is nonradial. Equation (4.1) then has at least two positive solutions, one radial and one nonradial. Similar symmetry breaking occurs for exponents near 2^* on a fixed annulus.

Exercise 4.4. Supply the estimates proving $m_{r,a} \rightarrow \infty$ and $\sup_a m_a < \infty$. Explain why distinct minimization classes give distinct positive solutions.

5 The Nehari method

We return to a bounded domain and consider

$$-\Delta u = f(u) \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega. \quad (5.1)$$

Let $E = H_0^1(\Omega)$ and $F(t) = \int_0^t f(s) \, ds$. Assume:

(F1) $f \in C^1(\mathbb{R}, \mathbb{R})$, $f(0) = f'(0) = 0$;

(F2) for some $2 < p < 2^*$, $|f'(t)| \leq C(1 + |t|^{p-2})$;

(F3) there are $\mu > 2$ and $M > 0$ such that $0 < \mu F(t) \leq t f(t)$ whenever $|t| \geq M$ (the Ambrosetti–Rabinowitz condition);

(F4) $f(t)/t$ is strictly increasing on each of the rays $(0, \infty)$ and $(-\infty, 0)$.

The model $f(t) = |t|^{p-2}t$ satisfies all four assumptions.

Define

$$I(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2 dx - \int_{\Omega} F(u) dx, \quad \mathcal{N} = \{u \in E \setminus \{0\} : \langle I'(u), u \rangle = 0\}. \quad (5.2)$$

The set $\mathcal{N}$ is the Nehari manifold. Since

$$\langle I'(u), u \rangle = \int_{\Omega} |\nabla u|^2 dx - \int_{\Omega} f(u)u dx,$$

every nonzero critical point lies in $\mathcal{N}$.

Lemma 5.1 (The Nehari set is a natural constraint). *Under (F1)–(F4), $\mathcal{N}$ is a C^1 manifold and every critical point of $I|_{\mathcal{N}}$ is a critical point of I in E .*

Proof. Let $G(u) = \langle I'(u), u \rangle$. For $u \in \mathcal{N}$,

$$\langle G'(u), u \rangle = \int_{\Omega} (f(u)u - f'(u)u^2) dx < 0$$

by (F4), so $G'(u) \neq 0$. If $I'(u) = \lambda G'(u)$ at a constrained critical point, testing with u gives $0 = \lambda \langle G'(u), u \rangle$, hence $\lambda = 0$. $\square$

Lemma 5.2 (Unique radial projection). *For every $u \in E \setminus \{0\}$ there is a unique $t(u) > 0$ such that $t(u)u \in \mathcal{N}$. Moreover $t \mapsto I(tu)$ has a unique global maximum at $t = t(u)$.*

Proof. For small t , (F2) and Sobolev embedding imply

$$I(tu) \geq c_1 t^2 \|u\|^2 - c_2 t^p \|u\|^p > 0,$$

whereas (F3) implies $I(tu) \rightarrow -\infty$ as $t \rightarrow \infty$. Thus a maximum exists. Put $g(t) = I(tu)$. At a critical point,

$$g''(t) = \frac{1}{t^2} \int_{\Omega} (f(tu)tu - f'(tu)(tu)^2) dx < 0,$$

so there is at most one critical point. Since $g'(t) = t^{-1} \langle I'(tu), tu \rangle$, that point is exactly the unique intersection with $\mathcal{N}$. $\square$

Define the ground level $m = \inf_{\mathcal{N}} I$. The small-ball estimate above gives $m > 0$. An $I|_{\mathcal{N}}$ -minimizing sequence is bounded: combine $I(u_n) \rightarrow m$ with $\langle I'(u_n), u_n \rangle = 0$ and (F3) to control the large-value part of u_n . Weak compactness, subcritical compactness, and the unique radial projection then give a minimizer.

Theorem 5.3 (Ground state on the Nehari manifold). *Under (F1)–(F4), $m = \inf_{\mathcal{N}} I$ is attained at some $u \in \mathcal{N}$. The minimizer is a critical point of I and has one sign. In particular, (5.1) has a positive or a negative ground state.*

Proof. Let u minimize on $\mathcal{N}$. Since $u^{\pm}$ have disjoint supports, $\langle I'(u), u^{\pm} \rangle = 0$. If both parts were nonzero, then $u^{\pm} \in \mathcal{N}$ and

$$m = I(u) = I(u^+) + I(u^-) \geq 2m,$$

contradicting $m > 0$. Thus u has one sign; the maximum principle gives a strictly positive or strictly negative solution. $\square$

The same Nehari construction applies to compact Schrödinger settings $-\Delta u + V(x)u = f(u)$, provided the potential restores compactness.

5.1 A formulation without the AR condition

The AR inequality is convenient but not necessary. Let $f(x, t)$ be a Carathéodory nonlinearity on a bounded domain and let $q = 2^*$ for $N \geq 3$ (choose any finite $q > 2$ when $N = 2$). Assume:

(H1) $|f(x, t)|/|t|^{q-1} \rightarrow 0$ as $|t| \rightarrow \infty$, uniformly in x ;

(H2) $f(x, t) = o(|t|)$ as $t \rightarrow 0$, uniformly in x ;

(H3) $F(x, t)/t^2 \rightarrow +\infty$ as $|t| \rightarrow \infty$, uniformly in x ;

(H4) $f(x, t)t - 2F(x, t)$ is nondecreasing in $|t|$ and strictly increasing near $t = 0$.

Set

$$\Phi(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2 dx - \int_{\Omega} F(x, u) dx.$$

Subcritical growth makes $\Phi \in C^1(E, \mathbb{R})$ and the map $t \mapsto \Phi(tu)$ has a unique positive maximizer. Therefore

$$c = \inf_{u \neq 0} \max_{t > 0} \Phi(tu) > 0$$

is attained at a nonzero critical point. The boundedness of a minimizing sequence follows by normalizing $v_n = u_n/\|u_n\|$: if the weak limit of v_n is nonzero, (H3) contradicts the bounded energy; if it is zero, testing at a fixed large radius contradicts the definition of c . Weak continuity of $\int F(x, u)$ and the unique-maximizer property finish the argument.

The same idea applied to positive and negative parts yields a least-energy sign-changing solution w with exactly two nodal domains and

$$\Phi(w) = \max_{t > 0} \Phi(tw^+) + \max_{t > 0} \Phi(tw^-).$$

6 Minimax preliminaries: deformation and compactness

6.1 Smoothness and pseudo-gradient vector fields

For a Banach space X , $I \in C^1(X, \mathbb{R})$, and $u \in X$, the directional derivative is

$$\langle I'(u), v \rangle = \lim_{t \rightarrow 0} \frac{I(u + tv) - I(u)}{t}.$$

On a Hilbert space, $I'(u)$ can be represented by $\nabla I(u)$ through $\langle I'(u), v \rangle = (\nabla I(u), v)$. For example, if $I(u) = \|u\|^2$ then $\nabla I(u) = 2u$. On a general Banach space there is no canonical gradient.

Definition 6.1 (Pseudo-gradient vector). If $I'(p) \neq 0$, a vector $V \in X$ is a pseudo-gradient vector at p if

$$\|V\| \leq 2 \|I'(p)\|_{X^*}, \text{ and } \langle I'(p), V \rangle \geq \|I'(p)\|_{X^*}^2.$$

Lemma 6.2. *At every noncritical point p a pseudo-gradient vector exists. Moreover, there is a locally Lipschitz vector field $V : X \setminus K \rightarrow X$, where $K = \{I' = 0\}$, whose value at each point is a pseudo-gradient vector.*

Proof. Choose x_1 with $\|x_1\| = 1$ and $\langle I'(p), x_1 \rangle \geq \frac{2}{3} \|I'(p)\|$. Then $V = \frac{3}{2} \|I'(p)\| x_1$ works. A locally finite partition of unity patches these local choices into a locally Lipschitz field. $\square$

The flow

$$\dot{\sigma}(t) = -V(\sigma(t)), \quad \sigma(0) = u,$$

satisfies

$$\frac{d}{dt}I(\sigma(t)) = -\langle I'(\sigma(t)), V(\sigma(t)) \rangle \leq -\|I'(\sigma(t))\|_{X^*}^2 \leq 0.$$

Thus pseudo-gradient flows decrease the functional without requiring C^2 smoothness.

6.2 The Palais–Smale condition

Definition 6.3 ((PS) $_c$). The functional $I \in C^1(X, \mathbb{R})$ satisfies the (PS) $_c$ condition if every sequence (u_n) such that

$$I(u_n) \rightarrow c, \quad \|I'(u_n)\|_{X^*} \rightarrow 0$$

has a convergent subsequence. It satisfies (PS) if this holds at every level.

If (PS) $_c$ holds, the critical set $K_c = \{u : I'(u) = 0, I(u) = c\}$ is compact. If (PS) holds on $[a, b]$ and there are no critical points with energies in that interval, then

$$\inf_{u \in I^{-1}([a, b])} \|I'(u)\|_{X^*} > 0.$$

This uniform bound is exactly what makes a deformation possible. Without compactness the conclusion fails, as shown by $I(x) = e^{-x}$ on $\mathbb{R}$: $I_a = \emptyset$ for $a \leq 0$, while $I_b = (-\infty, \log(1/b)]$ for $b > 0$, and the sublevel topology changes without a critical point.

Theorem 6.4 (First deformation lemma). *Let $I \in C^1(X, \mathbb{R})$ satisfy (PS) $_c$, let K_c be its critical set at level c , and let N be an open neighborhood of K_c . For sufficiently small $0 < \varepsilon < \bar{\varepsilon}$, there is a continuous deformation $\eta : [0, 1] \times X \rightarrow X$ such that:*

- (i) $\eta(0, u) = u$;
- (ii) $\eta(t, u) = u$ outside $I^{-1}([c - \bar{\varepsilon}, c + \bar{\varepsilon}])$;
- (iii) each $\eta(t, \cdot)$ is a homeomorphism and $I(\eta(t, u))$ is nonincreasing in t ;
- (iv) $\eta(1, I_{c+\varepsilon} \setminus N) \subset I_{c-\varepsilon}$;
- (v) if $K_c = \emptyset$, then $\eta(1, I_{c+\varepsilon}) \subset I_{c-\varepsilon}$.

The second deformation lemma says that if (PS) holds on $[a, b]$ and there are no critical points in $I^{-1}([a, b])$, then I_a is a strong deformation retract of I_b . A quantitative version does not require (PS): if for some set S and $\varepsilon, \delta > 0$,

$$\|I'(u)\| \geq \frac{8\varepsilon}{\delta} \quad \text{on } I^{-1}([c - 2\varepsilon, c + 2\varepsilon]) \cap S_{2\delta},$$

then a deformation lowers $I_{c+\varepsilon} \cap S$ into $I_{c-\varepsilon}$, moves points by at most δ , and leaves the complement of the indicated energy strip fixed. Construct it from a cut-off of the pseudo-gradient field:

$$\psi(u) = \frac{\text{dist}(u, X \setminus A)}{\text{dist}(u, X \setminus A) + \text{dist}(u, X \setminus B)}, \quad W(u) = -\psi(u) \frac{V(u)}{\|V(u)\|^2}.$$

7 Mountain-pass and genus methods

7.1 The mountain-pass theorem

Theorem 7.1 (Ambrosetti–Rabinowitz mountain pass). *Let E be a Banach space and $I \in C^1(E, \mathbb{R})$ satisfy (PS). Suppose*

(M1) $I(0) = 0$ and there are $\rho, \alpha > 0$ with $I(u) \geq \alpha$ whenever $\|u\| = \rho$;

(M2) there is $e \in E$ with $\|e\| > \rho$ and $I(e) \leq 0$.

Then I has a critical point at the level

$$c = \inf_{\gamma \in \Gamma} \max_{t \in [0,1]} I(\gamma(t)) \geq \alpha, \quad \Gamma = \{\gamma \in C([0,1], E) : \gamma(0) = 0, \gamma(1) = e\}.$$

Proof. The number c is finite and $c \geq \alpha$ because every path crosses the sphere of radius ρ . If c were not a critical value, the first deformation lemma would lower every path whose maximum is below $c + \varepsilon$ to one whose maximum is below $c - \varepsilon$, contradicting the definition of c . $\square$

7.2 Application to a superlinear Dirichlet problem

For

$$-\Delta u = g(u) \text{ in } \Omega, \quad u = 0 \text{ on } \partial\Omega,$$

let

$$J(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2 dx - \int_{\Omega} G(u) dx, \quad G(t) = \int_0^t g(s) ds.$$

Assume the AR hypotheses: $g \in C^1$, $g(0) = g'(0) = 0$; subcritical growth $|g(t)| \leq C(1 + |t|^{p-1})$ for $2 < p < 2^*$; and $0 < \mu G(t) \leq t g(t)$ for $|t|$ large and some $\mu > 2$. Then $J \in C^1$. The small- t estimate

$$|G(t)| \leq \frac{\varepsilon}{2} |t|^2 + C_{\varepsilon} |t|^p$$

gives, by Sobolev embedding,

$$J(u) \geq \frac{1}{4} \|u\|^2 - C \|u\|^p$$

for an appropriate choice of ε . Choose ρ small so the right side is positive on ∂B_{ρ} . The AR condition implies $J(tu) \rightarrow -\infty$ for every $u \neq 0$, giving the second mountain-pass condition. A bounded (PS) sequence has a weakly convergent subsequence; the subcritical embedding makes the nonlinear term converge strongly, hence the sequence converges in H_0^1 . The theorem gives a nonzero weak solution. By testing with the negative part and using truncation, one obtains a nonnegative solution, which is positive by the maximum principle. Applying the argument to $-g$ gives a negative solution when the hypotheses are symmetric.

7.3 Genus and Clark's theorem

Let E be a Banach space. For a closed symmetric set $A = -A$, its Krasnosel'skii genus $\gamma(A)$ is the smallest $k \in \mathbb{N}$ for which there is a continuous odd map $A \rightarrow S^{k-1}$; set $\gamma(A) = \infty$ if none exists. We use:

$$\begin{aligned} A \subset B &\Rightarrow \gamma(A) \leq \gamma(B), & \gamma(A \cup B) &\leq \gamma(A) + \gamma(B), \\ g : A &\rightarrow E \text{ continuous and odd} &\Rightarrow \gamma(A) &\leq \gamma(\overline{g(A)}), \end{aligned}$$

and $\gamma(S^{k-1}) = k$. The Borsuk–Ulam theorem is the topological input: every continuous odd map $S^{k-1} \rightarrow \mathbb{R}^{k-1}$ has a zero. Equivalently, an odd map from the unit ball to $\mathbb{R}^k$ that is odd on the boundary has a zero.

Theorem 7.2 (Clark). *Let $I \in C^1(X, \mathbb{R})$ be bounded below, even, $I(0) = 0$, and satisfy $(PS)_c$ for every $c < 0$. If there are a k -dimensional subspace X_k and $\delta > 0$ such that*

$$\sup_{x \in X_k, \|x\|=\delta} I(x) < 0,$$

then I has at least k pairs of critical points with negative critical values.

Proof sketch. Let $\Gamma_j = \{A : A = -A, A \text{ compact}, \gamma(A) \geq j\}$ and

$$c_j = \inf_{A \in \Gamma_j} \sup_{u \in A} I(u), \quad 1 \leq j \leq k.$$

The sphere of radius δ in X_k has genus k , so $c_1 \leq \dots \leq c_k < 0$. If a c_j were not critical, deformation would lower a set of genus at least j below c_j , a contradiction. If $c_j = \dots = c_{j+p}$, a deformation away from a small neighborhood of the critical set and the subadditivity of genus imply $\gamma(K_{c_j}) \geq p + 1$. Thus repeated levels still give sufficiently many critical points. $\square$

Corollary 7.3 (Infinitely many small negative solutions). *If for every k there is a k -dimensional subspace X_k and radius δ_k with $\sup_{\|x\|=\delta_k, x \in X_k} I(x) < 0$, then there are critical values $c_k < 0$ with $c_k \rightarrow 0$.*

Applications include

$$-\Delta u = \lambda |u|^{q-2} u, \quad 1 < q < 2,$$

which has infinitely many solutions with negative energy, and odd nonlinear pendulum problems such as $g(u) = a \sin u$ in one dimension. If $g'(0) > \lambda_k$ and $|g(u)| \leq C + \lambda |u|$ with $\lambda < \lambda_1$, Clark's theorem gives at least k pairs of solutions.

A refinement due to Liu–Wang states that under the same hypotheses either there is a sequence of negative-energy critical points converging to 0 in norm, or every sufficiently small sphere contains a zero-energy critical point. This alternative is useful when the zero level is not isolated.

7.4 Invariant cones and sign-changing solutions

Let $X = C_0^1(\overline{\Omega})$ and let $P = \{u \in X : u \geq 0\}$, with $-P$ the negative cone. For many elliptic functionals the negative gradient flow leaves P and $-P$ invariant and sends nonzero cone points into their interiors. Define minimax levels using

$$\sup_{A \setminus (P \cup (-P))} I.$$

If the level is critical, its critical point lies outside the two cones and is therefore sign-changing. This is the mechanism behind the existence of pairs of nodal solutions in the partially ordered Hilbert-space version of Ljusternik–Schnirelmann theory.

8 Symmetric mountain pass and saddle-point constructions

8.1 The symmetric mountain-pass theorem

Let E be infinite-dimensional, $I \in C^1(E, \mathbb{R})$ even, $I(0) = 0$, and assume (PS) . Suppose there are $\rho, \alpha > 0$ with $I \geq \alpha$ on ∂B_ρ , and for every n an n -dimensional subspace E_n and $R_n > \rho$ such that

$$I(u) \leq 0 \quad \text{for } u \in E_n \setminus B_{R_n}.$$

Theorem 8.1 (Symmetric mountain pass). *Under these assumptions, I has an unbounded sequence of critical values.*

For completeness, the minimax families are built from

$$G_n = \{\varphi \in C(B_{R_n}, E) : \varphi(-u) = -\varphi(u), \varphi|_{\partial B_{R_n}} = \text{Id}\}.$$

If $Y = -Y \subset B_{R_n}$ is open with $\gamma(\bar{Y}) \leq n - j$, set

$$\Gamma_j = \{\varphi(B_{R_n} \setminus Y) : \varphi \in G_n, n \geq j\}, \quad c_j = \inf_{A \in \Gamma_j} \sup_{u \in A} I(u).$$

Genus properties, deformation, and the intersection of every $A \in \Gamma_j$ with ∂B_ρ give $c_j \geq \alpha$ and show that each c_j is a critical value. If a level repeats, the critical set has large genus. Finally, to prove $c_j \rightarrow \infty$, let V be a fixed finite-dimensional subspace and write $E = V \oplus X$. For $j > \dim V$, every $A \in \Gamma_j$ intersects $\partial B_\rho \cap X$; applying the positive lower bound on larger and larger finite-dimensional subspaces forces the levels upward.

For the model power nonlinearity, this yields an unbounded sequence of bound states of

$$-\Delta u = |u|^{p-2}u \quad \text{in } \Omega, \quad u = 0 \text{ on } \partial\Omega, \quad 2 < p < 2^*.$$

8.2 A saddle-point alternative

Fix a j -dimensional ball $B_{R_j} \subset E$ and define

$$\Gamma_j = \{\varphi(B_{R_j}) : \varphi : B_{R_j} \rightarrow E \text{ continuous and odd, } \varphi|_{\partial B_{R_j}} = \text{Id}\}, \quad c_j = \inf_{A \in \Gamma_j} \sup_A I.$$

Under the corresponding linking geometry and a suitable (PS) condition, each c_j is critical. This formulation is often called the saddle-point theorem and can replace the symmetric mountain-pass construction when the negative directions are naturally finite-dimensional.

8.3 Linear eigenvalues as minimax levels

For the Dirichlet Laplacian on a bounded domain,

$$\lambda_1 = \inf_{\|u\|_2=1} \|\nabla u\|_2^2,$$

and, inductively,

$$\lambda_k = \inf \left\{ \|\nabla u\|_2^2 : \|u\|_2 = 1, \int_{\Omega} u \phi_i \, dx = 0 \ (1 \leq i < k) \right\}.$$

The genus minimax values recover the same spectrum and explain why the first eigenfunction is one-sign while higher eigenfunctions are nodal.

9 Quasilinear problems and variational perturbation

9.1 A general quasilinear variational equation

Let $a_{ij} = a_{ji}$ be C^1 in the scalar variable and let $F(x, s)$ be a primitive of $f(x, s)$. Formally set

$$J_0(u) = \frac{1}{2} \int_{\Omega} \left(\sum_{i,j=1}^N a_{ij}(x, u) \partial_i u \partial_j u + V(x) u^2 \right) dx - \int_{\Omega} F(x, u) dx. \quad (9.1)$$

Its Euler equation is

$$\begin{aligned} \int_{\Omega} \sum_{i,j} a_{ij}(x, u) \partial_i u \partial_j \phi \, dx + \frac{1}{2} \int_{\Omega} \sum_{i,j} \partial_s a_{ij}(x, u) \partial_i u \partial_j u \phi \, dx \\ + \int_{\Omega} V(x) u \phi \, dx - \int_{\Omega} f(x, u) \phi \, dx = 0 \end{aligned} \quad (9.2)$$

for $\phi \in C_c^\infty(\Omega)$. The natural weak-solution class is often $H_0^1(\Omega) \cap L^\infty(\Omega)$. The dependence of a_{ij} on u changes both existence and critical exponents.

For the modified nonlinear Schrödinger equation (MNLS), take

$$a_{ij}(x, s) = (1 + s^2) \delta_{ij}.$$

Then (9.2) becomes

$$\int_{\Omega} (1 + u^2) \nabla u \cdot \nabla \phi + u |\nabla u|^2 \phi + V(x) u \phi - f(x, u) \phi \, dx = 0. \quad (9.3)$$

Equivalently,

$$-\operatorname{div}((1 + u^2) \nabla u) + u |\nabla u|^2 + V(x) u - f(x, u) = 0, \quad (9.4)$$

or

$$-\Delta u - \frac{1}{2} u \Delta(u^2) + V(x) u - f(x, u) = 0.$$

The formal energy is

$$J_0(u) = \frac{1}{2} \int_{\mathbb{R}^N} ((1 + u^2) |\nabla u|^2 + V(x) u^2) \, dx - \int_{\mathbb{R}^N} F(x, u) \, dx. \quad (9.5)$$

For $N \geq 2$ the term $\int u^2 |\nabla u|^2$ need not be finite on $H^1(\mathbb{R}^N)$, so J_0 is not a C^1 functional on the usual Sobolev space. This is the central analytic difficulty.

When $a_{ij}(x, s) = a(s) \delta_{ij}$, the change of variable

$$v = h(u) := \int_0^u \sqrt{1 + s^2} \, ds = \frac{1}{2} u \sqrt{1 + u^2} + \frac{1}{2} \operatorname{arsinh} u \quad (9.6)$$

has inverse $u = g(v)$ and transforms MNLS formally into

$$-\Delta v + V(x) g(v) g'(v) = f(x, g(v)) g'(v).$$

This reduces the principal quasilinear term to a semilinear one, but the transformed nonlinearity must still be controlled at the correct growth rate.

For general matrices, useful structural assumptions are:

$$c_0^{-1} (1 + s^2) |\xi|^2 \leq \sum_{i,j} a_{ij}(s) \xi_i \xi_j \leq c_0 (1 + s^2) |\xi|^2, \quad (9.7)$$

$$\beta \sum_{i,j} a_{ij}(s) \xi_i \xi_j \leq \sum_{i,j} \left(a_{ij}(s) + \frac{1}{2} s a'_{ij}(s) \right) \xi_i \xi_j \leq \left(\frac{q}{2} - \beta \right) \sum_{i,j} a_{ij}(s) \xi_i \xi_j, \quad (9.8)$$

$$a_{ij}(-s) = a_{ij}(s). \quad (9.9)$$

Here $\beta > 0$ and the exponent lies in the quasilinear subcritical range $4 < q < 4N/(N - 2)$ for $N \geq 3$. A frequently used simpler version of (9.8) is

$$0 \leq \sum_{i,j} s a'_{ij}(s) \xi_i \xi_j \leq 2 \sum_{i,j} a_{ij}(s) \xi_i \xi_j.$$

Under these conditions, Nehari and nodal Nehari minimization can produce ground states and least-energy sign-changing states when the potential restores compactness.

9.2 The quasilinear critical exponent

The natural critical exponent for MNLS is

$$2_{\text{ql}}^* := 2 \cdot 2^* = \frac{4N}{N-2} \quad (N \geq 3).$$

A typical constrained quantity is

$$\inf_{\|u\|_{L^q}=1} \int_{\mathbb{R}^N} ((1+u^2)|\nabla u|^2 + V(x)u^2) \, dx.$$

Minimizers exist for $2 < q < 2_{\text{ql}}^*$ when V is coercive or radial, and for $4 \leq q < 2_{\text{ql}}^*$ in several periodic or potential-well settings. The multiplier equation is

$$-\Delta u - \frac{1}{2}u\Delta(u^2) + V(x)u = \lambda|u|^{q-2}u.$$

For $4 \leq q < 2_{\text{ql}}^*$, sequences of multipliers can approach both zero and infinity; a direct rescaling no longer removes λ . This is why a Nehari construction adapted to the quasilinear energy is useful. Under conditions such as

$$V(x) \leq V_\infty, \quad V(x) \leq V_\infty - \frac{A}{1+|x|^m} \quad (|x| > M),$$

one obtains respectively a positive ground state and a sign-changing weak solution in the range $4 \leq q < 2_{\text{ql}}^*$.

9.3 A variational perturbation

To recover smoothness, work in $W_0^{1,4}(\Omega)$ and define

$$J_\mu(u) = \frac{\mu}{4} \int_{\Omega} |\nabla u|^4 \, dx + J_0(u), \quad \mu > 0. \quad (9.10)$$

The fourth-order regularization makes J_μ differentiable. A typical convergence theorem is the following.

Theorem 9.1 (Perturbation convergence). *Assume (9.7)–(9.9) and suitable subcritical growth and regularity assumptions on f . Let $\mu_n \downarrow 0$ and $u_n \in W_0^{1,4}(\Omega)$ satisfy $J'_{\mu_n}(u_n) = 0$ and $J_{\mu_n}(u_n) \leq C$. Then, after a subsequence,*

$$\begin{aligned} \mu_n \|u_n\|_{W_0^{1,4}}^4 &\rightarrow 0, & u_n &\rightarrow u \text{ strongly in } W_0^{1,2}(\Omega), \\ u_n \nabla u_n &\rightarrow u \nabla u \text{ strongly in } L^2(\Omega), & J_{\mu_n}(u_n) &\rightarrow J_0(u), \end{aligned}$$

and u is a weak solution of the unregularized quasilinear equation.

Proof architecture

The proof in the lecture notes consists of four reusable steps.

1. Uniform bounds. Test $J'_{\mu_n}(u_n) = 0$ with u_n and combine the result with the energy bound. The structural inequalities imply

$$\mu_n \int_{\Omega} |\nabla u_n|^4 \, dx \leq C, \quad \int_{\Omega} \sum_{i,j} a_{ij}(u_n) \partial_i u_n \partial_j u_n \, dx \leq C.$$

Hence, up to a subsequence,

$$u_n \rightharpoonup u \text{ in } W_0^{1,2}, \quad u_n \nabla u_n \rightharpoonup u \nabla u \text{ in } L^2,$$

and $u_n \rightarrow u$ in L_{loc}^r for every $2 \leq r < 2_{\text{ql}}^*$.

2. Uniform L^∞ control. Let u_n^T be the truncation at height T and test with $\phi = |u_n^T|^{2r} u_n^T$. Sobolev's inequality gives the iteration estimate

$$\|u_n\|_{L^{(2+r)2^*/2}} \leq \left(Cr \int_{\Omega} |u_n|^{q+2r} dx \right)^{1/[2(2+r)]}.$$

Choosing a sequence $r_{k+1} = (2^*/2)(2 + r_k) - 2$ and iterating gives $\|u_n\|_{L^\infty} \leq C$ independently of n .

3. Passing to the weak equation. Choose $T > \|u\|_{L^\infty}$ and define

$$w_n = \max\{u_n, -T\}.$$

For $\psi \geq 0$ in $C_c^\infty(\Omega)$ use $\phi = \psi e^{-Mw_n}$. The resulting identity contains the coercive term

$$\int_{\{u_n \geq -T\}} \left(Ma_{ij}(u_n) - \frac{1}{2} a'_{ij}(u_n) \right) \partial_i u_n \partial_j u_n \psi e^{-Mw_n} dx.$$

Choose M so the matrix in parentheses is positive. The regularizing error satisfies

$$\mu_n \int |\nabla u_n|^2 |\nabla \psi| e^{-Mw_n} dx \leq C \mu_n^{1/4} \rightarrow 0.$$

Lower semicontinuity and local convergence give, for nonnegative $\phi \in C_c^\infty(\Omega)$,

$$\int a_{ij}(u) \partial_i u \partial_j \phi dx + \frac{1}{2} \int a'_{ij}(u) \partial_i u \partial_j u \phi dx \geq \int f(x, u) \phi dx.$$

Using $v_n = \min\{u_n, T\}$ and e^{Mv_n} gives the reverse inequality. Approximation extends the identity to all bounded H_0^1 test functions.

4. Strong convergence. Test the equations with u_n and u , respectively:

$$\begin{aligned} \mu_n \int |\nabla u_n|^4 + \int a_{ij}(u_n) \partial_i u_n \partial_j u_n + \frac{1}{2} \int a'_{ij}(u_n) u_n \partial_i u_n \partial_j u_n &= \int f(x, u_n) u_n, \\ \int a_{ij}(u) \partial_i u \partial_j u + \frac{1}{2} \int a'_{ij}(u) u \partial_i u \partial_j u &= \int f(x, u) u. \end{aligned}$$

The convergence of the right-hand sides and the monotonicity supplied by (9.7)–(9.8) imply $\mu_n \|u_n\|_{W^{1,4}}^4 \rightarrow 0$, strong convergence in H_0^1 , and $u_n \nabla u_n \rightarrow u \nabla u$ in L^2 .

The regularized functionals have infinitely many critical values in the appropriate symmetric subcritical range. The convergence theorem transfers the limiting sequence to the original quasilinear problem, producing an unbounded family of weak solutions.

9.4 Pohozaev identities and nonexistence

For comparison, if $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ solves $-\Delta u = f(u)$ with $u = 0$ on $\partial\Omega$, then

$$\frac{N-2}{2} \int_{\Omega} |\nabla u|^2 dx - N \int_{\Omega} F(u) dx + \frac{1}{2} \int_{\partial\Omega} (x \cdot \nu) \left| \frac{\partial u}{\partial \nu} \right|^2 d\sigma = 0, \quad (9.11)$$

where ν is the outward unit normal and $F(t) = \int_0^t f(s) ds$. For quasilinear equations, Pucci–Serrin type identities contain additional terms from the coefficient $a_{ij}(u)$. In the MNLS model they show that the threshold $q = 4N/(N-2)$ is critical: under star-shaped boundary conditions, solutions are generally ruled out at and above this exponent.